

Feasibility of Embedded UAV-Based Semantic Segmentation for Lichen Detection in the Namib Desert

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Abstract - Seeking to tackle some of the key inefficiencies of the search for life on Mars through analogs, this study investigates the feasibility of deploying lightweight deep learning semantic segmentation models for autonomous lichen detection in Martian analog environments using UAV edge-computing hardware. Lichen was selected as a proxy for extraterrestrial life due to its demonstrated resilience under Martian environmental conditions, while the Namib Desert was chosen as an environmental analog for remote planetary exploration. U-Net and DeepLabV3+ architectures paired with ResNet18, ResNet34, and MobileNetV2 encoder backbones were trained on manually annotated RGB imagery and deployed on an NVIDIA Jetson Nano to evaluate segmentation performance and inference latency under resource-constrained conditions. A “Next-Step Proxy” metric was additionally developed to simulate autonomous UAV navigation decisions by identifying image regions with the highest predicted lichen density. Results showed that DeepLabV3+ with a MobileNetV2 backbone achieved the strongest overall balance of segmentation accuracy and computational efficiency, producing the fastest steady-state inference latency among all evaluated configurations. A human-in-the-loop dataset expansion pipeline improved large-scale spatial inference performance but reduced conventional pixel-level segmentation metrics, likely due to increased annotation variability. These findings demonstrate the potential of lightweight semantic segmentation systems for autonomous ecological surveying.

I. INTRODUCTION

A. Background

The search for signs of extraterrestrial life on the Martian surface has many layers of logistical and operational challenges. While the detection of biosignatures by deployed rovers continues to garner public excitement, the speculative nature of the forms life could or could have taken in the evolving Martian environment has left the field of astrobiology far from consensus (Taveau 2025). In addition to unclear search parameters or objectives, the difficult Martian terrain, coupled with the 12-minute communication delay to Earth, makes the operations of the Mars

rovers incredibly gradual and inherently inefficient (Gaines et al. 2016).

In order to circumvent the terrain challenges of ground-based Martian exploration, NASA has shifted to the deployment of Unmanned Aerial Vehicles (UAVs). In 2021, the Ingenuity Mars Helicopter began conducting aerial missions on the surface of the Martian surface, which lasted three years (NASA 2024). With an atmosphere that is only 1% of Earth's atmospheric density, Martian UAV deployment is inherently payload-limited (Brown 2026). Therefore, the engineering question of these proposed deployments is how to collect relevant data as efficiently as possible with the limited on-board power and computing resources.

One area of interest for UAVs is autonomous navigation and control decisions via deep-learning-empowered computer vision interfaces. Computer vision systems allow UAVs to identify obstacles, estimate depth, classify terrain, and make navigation decisions using convolutional neural networks (CNNs) and transformer-based architectures (Li et al. 2024). It follows that such methods would be useful for UAVs to make autonomous path decisions, prioritizing areas of interest when deployed on the Martian surface.

When developing technology for such remote deployment, it is useful to identify both environmental proxies for the Martian surface and biological proxies for extraterrestrial life. Lichen has survived and photosynthesized under laboratory-simulated Martian polar conditions, as well as in the vacuum of space, and is therefore considered a promising candidate for Martian analog organisms (de Vera et al. 2004, de Vera et al. 2009). Deserts such as the Namib are among the closest environments to the

Martian surface on the planet and are therefore sites of methodological and technological development for Martian deployment (Hinchliffe et al. 2017). While not focused on autonomous navigation or embedded deep learning, myriad UAV-based studies in deserts such as Antarctica, the Atacama, and the Namib have deployed deep learning models to identify lichen (Hinchliffe et al. 2017, Amarasingam et al. 2025).

Existing research on DL applications for UAV classification of lichen in desert conditions, with a focus on martian analogs, exists but has focused on classifying lichen from UAV-generated datasets rather than on autonomous operation or edge computing. The two most successfully applied Deep Learning models for lichen identification have been U-Net and DeepLabV3+ (Amarasingam et al. 2025). Both are convolutional neural network architectures designed specifically for semantic segmentation, with pixel-wise classification outputs. Both architectures have two main components, an encoder and a decoder. The encoder progressively compresses the input image and extracts abstract spatial features, while the decoder reconstructs the image using a segmentation map. For the U-Net, skip connections between the corresponding encoder-decoder layers preserve details that would otherwise be lost during downsampling, which is beneficial for preserving the distinctive characteristic texture of lichen (Ronneberger et al., 2015). DeepLabV3+ has less dependence on progressive image compression, Atrous Spatial Pyramid Pooling (ASPP), in which dilated (“atrous”) convolutions expand the receptive field without significantly increasing the number of parameters or reducing image resolution. This enables the model to capture information at multiple spatial scales simultaneously (Chen et al. 2017). DeepLabV3+ generally achieves superior performance on complex semantic segmentation tasks but is more architecturally

complex and therefore computationally more expensive than U-Net.

B. Problem Statement and Motivation

Autonomous planetary exploration faces major operational constraints due to hazardous terrain, communication delays, and limited onboard computational resources. While UAVs such as NASA’s Ingenuity helicopter offer a promising alternative to slow ground-based rover traversal, their strict payload and power limitations require lightweight, computationally efficient onboard perception systems capable of operating with minimal human intervention. Existing deep learning approaches for lichen identification in Martian analog environments have primarily focused on offline analysis of UAV imagery rather than real-time edge deployment or autonomous navigation support. This project addresses the problem of whether lightweight semantic segmentation architectures can accurately identify lichen—a promising biological analog for extraterrestrial life—while maintaining inference speeds suitable for deployment on resource-constrained UAV edge devices operating in remote or extraterrestrial environments.

C. Objectives & Contribution

The aim of this project is to evaluate whether lightweight deep learning semantic segmentation models could enable autonomous, edge-computing-based identification of lichen in Martian analog environments using computationally constrained UAV platforms using a JetsonNano as a proxy for a computationally limited environment. Seeking to compare U-Net and DeepLabV3+ segmentation architectures with ResNet18, ResNet34, and MobileNetV2 encoder backbones deployed on a Jetson Nano edge device. It also seeks to iron out other practical pipeline concerns for

conducting field projects, such as implications of model latency, human-in-the-loop dataset expansion, and application-specific navigation proxies for autonomous scientific exploration.

II. RELATED WORKS

Applications of these deep learning models on UAV edge devices have been successful in other remote sensing domains. One of the architectural differences between these applications has been the encoder choice for computational efficiency. On edge-computing devices, especially with limited power and payload like UAVs, more lightweight models are needed than when lichen identification and analysis are being accomplished on traditional compute resources. For edge computing scenarios, encoder backbones such as ResNet-18 and MobileNetV2 are frequently chosen because they strike a balance between feature-extraction performance and computational efficiency. ResNet-based encoders preserve strong representational capacity through residual connections, allowing models to capture complex spatial textures and boundaries while remaining relatively lightweight in smaller variants such as ResNet-18. In contrast, MobileNetV2 is specifically optimized for edge deployment through depthwise separable convolutions and inverted residual blocks, substantially reducing parameter count, memory usage, and power consumption while maintaining acceptable segmentation accuracy. These characteristics make both backbone families well-suited for UAV-based semantic segmentation tasks where real-time inference and limited onboard compute resources are major constraints (Lai et al 2025).

In agricultural remote sensing, U-Net models with ResNet Encoders have been effectively deployed on UAVs. In a study assessing agricultural boundaries using UAV imaging, this hybrid framework effectively delineated field boundaries with 81% accuracy

and an inference time of 1ms (Li et al. 2023, Sarmah et al. 2025). MobileV2-UNet models had similar applications and success in agricultural boundary detection aboard UAV edge devices, and an average IoU of 0.908 (He et al. 2022). Deeplabv3+ models with a MobileNetV2 Encoder Backbone have been used to identify bare-rock geological features from UAV surveys with 97.39% accuracy (Lai et al. 2025). Deeplabv3+ with a ResNet-18 encoder has been shown to outperform MobileNetV2 in the UAV image analysis context (Sundaresan and Solomon 2025). Both geological classification and agricultural boundaries are near-enough classification tasks to have promising cross-applications for lichen density.

III. METHODOLOGY AND IMPLEMENTATION DETAILS

A. Training Data Source

Due to the unavailability of UAV datasets, the data were sourced from the iNaturalist database, sorting for images of lichen in Namibia. This dataset consisted of images with a range of zoom levels, resolutions, and lighting conditions. This was not ideal for the intended navigation objective, but was an acceptable preliminary step due to the qualitative similarities between lichen's growth on a micro and macro scale.



Image 1: Example dataset image from iNaturalist. *Namib Sun Lichen (Stellarangia namibensis)* (bhkisisa 2011).

One of the challenges of this dataset is that it consists of RGB images instead of the multispectral images of lichen that are characteristic of lichen UAV surveys (Hinchliffe et al. 2017, Amarasingam et al. 2025). Near-infrared radiation is characteristic of vegetative indices that have been useful to identify lichen. RGB pipelines are limited in the usage of wavelength masks for lichen identification. This perceived limitation is advantageous for edge computing for two reasons. One being the computational complexity necessary to parse the high-dimensional data of multispectral imagery, which is less feasible with a limited UAV payload and less advantageous for a low-latency objective (Amarasingam et al. 2025). Secondly, while existing applications of Deep Learning to UAV lichen images exist, they are not for edge computing and for post-image collection lichen species classification, for which color is a primary differentiator. Since the objective of this analysis is binary (presence/absence of lichen), it is less consequential to have multispectral data. The minute and complex growth patterns of lichen are perhaps more effectively assessed by a lightweight deep learning model than would be possible with input from a multispectral UAV.

B. Overall System/Model Architecture

For the purpose of this experiment, two semantic segmentation architectures, U-Net and DeepLabV3+, were evaluated using ResNet-18, ResNet-34, and MobileNetV2 encoder backbones. These encoder architectures were selected to compare the tradeoff between segmentation performance and computational efficiency for deployment on resource-constrained UAV edge devices. Each model produced pixel-wise binary segmentation masks indicating the presence or absence of lichen within an input image.

The trained models were deployed on an NVIDIA Jetson Nano edge-computing platform to simulate a lightweight onboard UAV inference environment. Saved model weights were loaded onto the Jetson Nano for inference on a held-out test dataset. During deployment, both segmentation performance metrics and inference latency measurements were collected. In addition to segmentation masks, a navigation-oriented “Next-Step Proxy” output was generated for each prediction to approximate autonomous UAV decision-making behavior.

C. Image Annotation and Dataset Preparation

Manual pixel-wise annotation for the binary segmentation task was conducted using the Computer Vision Annotation Tool (CVAT), an open-source image annotation platform. The resulting binary masks were exported and divided into training, validation, and test datasets.

D. Segmetic Segmentation Models

Two semantic segmentation frameworks, U-Net and DeepLabV3+ were evaluated. Each were paired with three encoder backbones: ResNet-18, ResNet-34 and MobileNetV2. These combinations were selected to evaluate the influence of encoder depth and computational complexity on segmentation accuracy and edge inference performance.

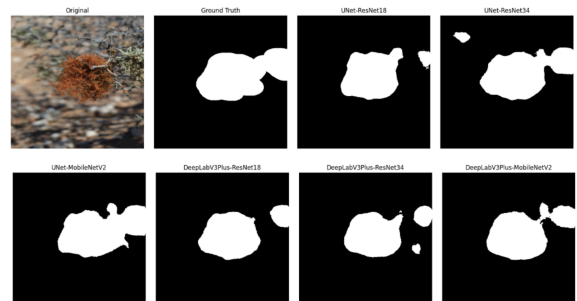


Figure 1: Example output of predictive masks generated by each model for an individual image in the training dataset.

E. Next-Step Proxy Module

Because the dataset was not geospatially referenced UAV imagery, a proxy navigation decision system was developed to simulate autonomous UAV path selection behavior. For each segmentation mask, the image was divided into four quadrants, and the number of lichen pixels within each quadrant was counted. The quadrant containing the highest lichen density was selected as the “Next-Step Proxy,” representing the hypothetical next navigation target for the UAV.

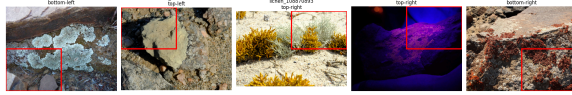


Figure 2: A visual representation of “Next-Step Proxy” overlaid on images from the dataset.

F. Human-in-the-Loop (HITL) Dataset Expansion

An additional training pipeline incorporating human-in-the-loop (HITL) annotation refinement was also evaluated. In this process, a U-Net model with a ResNet-34 encoder was initially trained on 150 manually labeled images. The trained model was then used to generate segmentation masks for 200 additional unlabeled images. These predicted masks were imported into CVAT and manually corrected to produce supplementary training annotations. The expanded dataset was subsequently used to retrain and evaluate the segmentation architectures.

G. Data Flow

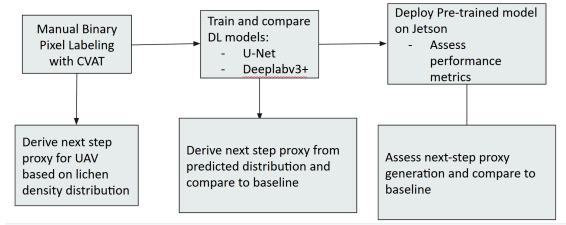


Figure 3: Original pipeline without HITL augmentation.

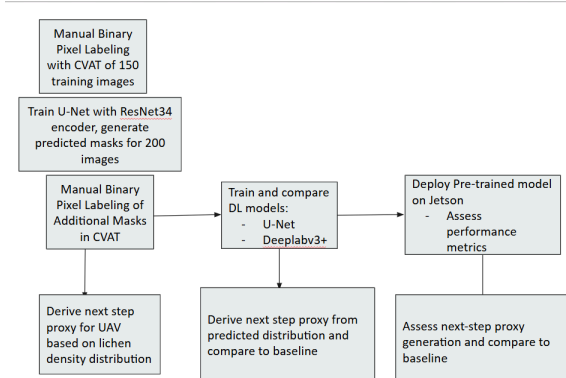


Figure 4: HITL pipeline.

IV: RESULTS AND EVALUATION

From each pre-trained model with training data comprised of only manual annotations, the following metrics were obtained from a test dataset.

Test Results					
	IoU	Dice	Precision	Recall	Accuracy
U-Net ResNet34	0.629715	0.772792	0.863632	0.699243	0.837034
U-Net ResNet18	0.665099	0.798870	0.864382	0.742589	0.851797
U-Net MobileNetV2	0.675105	0.806045	0.836226	0.777968	0.851608
DeepLabV3+ ResNet34	0.667802	0.800817	0.858097	0.750705	0.851988
DeepLabV3+ ResNet18	0.641641	0.781707	0.858548	0.717490	0.841174
DeepLabV3+ MobileNetV2	0.683116	0.811728	0.814504	0.808971	0.851264

Table 1: Demonstrates the metrics computed for the test data of each model with training data/masks pairs manually generated using CVAT. For the above pipeline, there were 191 Training images, 41 validation images, and 42 test images. The data followed a very clear pattern with respect to IoU, which is a particularly relevant performance metric for this task, with DeepLabV3+ performing better than U-Net, and MobileNetV2 performing better than ResNet34, which performed better than ResNet18.

From each model with the training data from HITL procedure, the following metrics were obtained from a test dataset.

Test Results

	IoU	Dice	Precision	Recall	Accuracy
U-Net ResNet34	0.622979	0.767698	0.692984	0.860470	0.830383
U-Net ResNet18	0.604969	0.753870	0.737335	0.771163	0.835984
U-Net MobileNetV2	0.607268	0.755652	0.747751	0.763722	0.839123
DeepLabV3+ ResNet34	0.626938	0.770697	0.771761	0.769635	0.850829
DeepLabV3+ ResNet18	0.588784	0.741176	0.744795	0.737591	0.832208
DeepLabV3+ MobileNetV2	0.625890	0.769904	0.743194	0.798606	0.844520

Table 2: Demonstrates the metrics computed for the test data of each model with training data/masks pairs produced from the HITL procedure. The performance was slightly worse and followed a different IoU performance pattern. For MobileNetV2 and 34 encoders, DeepLabV3+ still outperformed U-Net. However, U-Net with ResNet18 outperformed DeepLabV3+ ResNet18. The best performing encoders for both models were ResNet34, followed by MobileNetV2, and then ResNet18.

For both above methodologies, the percent alignment of the “Next-Step Proxy” from predicted masks and the “Next-Step Proxy” of the manually generated masks for the test dataset were as follows.

“Next-Step Proxy”	Manual Pixle Labeling	HITL
U-Net ResNet34	0.5238	0.6
U-Net ResNet18	0.5952	0.64342
U-Net MobileNetV2	0.5238	0.63030
DeepLabV3+ ResNet34	0.4762	0.70303
DeepLabV3+ ResNet18	0.5238	0.63030
DeepLabV3+ MobileNetV2	0.5952	0.63030

Table 3: The above table demonstrates the accuracy of “Next-Step Proxy” as determined by the predictive masks for both the Manual and HITL datasets. HITL seems to have significantly higher “Next-Step Proxy” accuracy outputs than the Manual.

The Average and Steady State Latency obtained from the deployment of a pretrained model trained on HITL on the test dataset is displayed below.

Model	Average Latency (s)	Steady State Latency (s)
U-Net ResNet34	2.106674	0.444319
U-Net ResNet18	2.187347	0.345305
U-Net MobileNetV2	2.084945	0.345191
DeepLabV3+ ResNet34	2.055530	0.506341
DeepLabV3+ ResNet18	2.013861	0.337390
DeepLabV3+ MobileNetV2	1.268814	0.302891

Table 4: Steady-state inference latency comparison for U-Net and DeepLabV3+ segmentation models using different encoder backbones. Reported values represent the average latency over the middle 10 inference runs to exclude initialization and GPU warmup overhead. Lower latency indicates faster inference performance.

V: DISCUSSION

DeepLabV3+ with a MobileNetV2 backbone achieved the strongest overall segmentation performance, suggesting that lightweight encoder architectures can effectively extract discriminative lichen features from UAV imagery while maintaining computational efficiency suitable for edge deployment. Steady-state inference latency measurements further supported this finding, with DeepLabV3+ MobileNetV2 producing the fastest inference times among all evaluated architectures. This was contrary to the expected computational cost of the models, as U-Net was initially regarded as a more lightweight model and therefore the most well-suited to

payload-limited edge computing applications. The lower comparable inference latency of DeepLabV3+ when compared to U-Net across encoders suggests that practical runtime performance is influenced not only by theoretical model complexity, but also by decoder efficiency, feature-map resolution, and GPU memory access patterns. Additionally, the limited improvement observed with deeper ResNet34 encoders suggests diminishing returns from increased model complexity, potentially due to dataset size limitations or the relatively moderate feature complexity of the imagery. Higher recall values in MobileNetV2-based models further indicate increased sensitivity to lichen presence, though with slightly reduced precision, reflecting a trade-off between false-positive suppression and detection of all lichen present.

The human-in-the-loop (HITL) training data expansion process resulted in lower overall segmentation performance across nearly all model architectures, particularly in IoU, Dice score, and precision metrics. This decline likely reflects the introduction of increased annotation variability within the expanded dataset. AI-assisted mask generation followed by manual correction may have introduced subtle inconsistencies in lichen boundary definition, contributing to elevated false positive rates and reduced precision, and produced polygon-based masks that are notoriously ill-suited to the complex growth structure of lichen (Amarasingam et al. 2025). At the same time, recall values remained comparatively high, suggesting that models retained sensitivity to lichen presence while becoming less selective in class assignment. Unlike the original dataset, where lightweight MobileNetV2 encoders performed second best overall, the expanded dataset favored DeepLabV3+ with a ResNet34 backbone, indicating that increased dataset complexity benefits from deeper contextual

feature extraction despite increased computational cost.

Despite reductions in conventional pixel-level segmentation metrics following human-in-the-loop dataset expansion, correct identification of the “Next-Step Proxy” or image quadrant containing the highest lichen density increased substantially across nearly all architectures. This suggests that the HITL-expanded dataset improved the models’ ability to capture large-scale spatial patterns and relative lichen abundance, even while introducing greater boundary-level ambiguity that negatively affected IoU and Dice scores. These findings indicate that conventional segmentation metrics may not fully represent ecological task performance in UAV-based lichen mapping applications. While fine-scale mask precision decreased, the models demonstrated improved ecological inference capability by more accurately identifying dominant lichen distribution regions. This distinction highlights the importance of evaluating semantic segmentation models using both pixel-level and application-specific spatial metrics in ecological remote sensing workflows.

VII: FUTURE WORK

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